Higher National Diploma in Information Technology
First Year, Second Semester Examination-2020
HNDIT1211: Data Structures and Algorithms

Instructions for Candidates:
Answer four (04) questions only.

No. of questions : 05
No. of pages : 06
Time : Two hours

Question 01

I. Define abstract data types. (2 marks)

II. State whether the following statements are correct or incorrect

a. Data structure reduces complexity of the program and its calculations.

b. An algorithm is a step-by-step procedure for solving a problem in an infinite amount of time.

c. According to 'Big O notation' O (1) is faster than O(n).

d. Best case efficiency is the Maximum number of steps that an algorithm can take any collection of data values (4 marks)

III. Write two differences between linear data structure and nonlinear data structures (4 marks)

IV. Write C++ function to reverse the inserted string using a for loop.
Note: Reverse string of "Hello" would be "olleH" (7 marks)

V. Write C++ function to display the subtraction of the matrices A and B given below.

$A=\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$ $B=\begin{bmatrix}10 & 20\\30 & 40\end{bmatrix}$

Note: Subtraction of A and B is A-B. (8 marks)

[Total 25 marks]

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Question 02

I. State two differences between array and Linked List. (4 marks)
II. Write two applications of Linked List (2 marks)
III. Mention a difference between singly linked list and doubly linked list. (2 marks)
IV. Graphically illustrate the static based (array based) implantation of the stack operations given below. Consider the size of the array is 5.

a) initializeStack()
b) x=isEmpty()
c) push(100); push(200)
d) y=pop()
e) r=isFull()
f) m=pop()
g) t=isEmpty()

(7 marks)

V. The following C++ class represents the implementation of several stack operations. Use that class to answer the questions given below.

```
#include<iostream.h>
#include<conio.h>
const STK_SIZE=5;
class Stack
{

private:
 int top;
 int stk[STK_SIZE];

public:
 Stack();
 void initializeStack();
 void push(int);
 int pop();
}
Stack::Stack()
{
 top=(-1);
}
void Stack::initializeStack()
{
 top=(-1);
}
```


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a) Write C++ code segment to implement the **push(int)** method in order to insert a value to the stack. (5 marks)

b) Write C++ code segment to implement the **pop()** method in order to remove a value from the stack. (5 marks)

[Total 25 marks]

Question 03

I. State two implementation of queue data structure. (2 marks)

II. Graphically illustrate the queue operations given below.

a) initializeQueue()
b) enQueue(100)
c) x=deQueue()
d) enQueue(200); enQueue(300); enQueue(400);
e) y=isEmpty()
f) displayQueue() (6 marks)

III. Graphically illustrate the static based (array based) implantation of the queue operations given below. Consider the size of the array is 5.

a) initializeQueue()
b) enQueue(1); enQueue(2); enQueue(3); enQueue(4)
c) x=deQueue()
d) enQueue(5); enQueue(6);
e) x=isFull() (8 marks)

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IV. The following C++ class represents the implementation of several queue operations. Use that class to answer the questions given below.

```
#include<iostream.h>
#include<conio.h>
const Q_SIZE=5;
class Queue
{
private:
 int front, rear, size;
 int que[Q_SIZE];
public:
 Queue();
 void initializeQueue();
 void enQueue(int);
 int deQueue();
}
Queue::Queue()
{
 front=(-1);
 rear=(-1);
 size=0;
}
void Queue::initializeQueue()
{
 front=(-1);
 rear=(-1);
 size=0;
}
```

a) Write C++ code segment to insert a value to the queue using **enQueue()** method. (4 marks)

b) Write C++ code segment to insert a value to the queue using **deQueue()** method. (5 marks)

[Total 25 marks]

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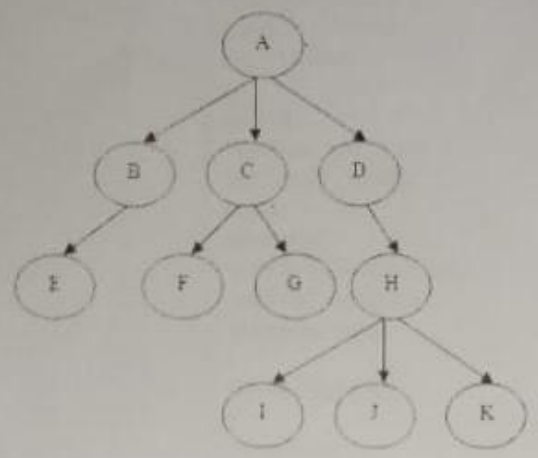
Question 4

I. Define Tree data structure. (2 marks)

II. Consider the following tree data structure to answer the questions given below.



a) What would be the size and height of the tree?
b) What is the depth of node H?
c) What is the degree of the tree and degree of node C?
d) State the path from root to node K
e) Mention the level of node F (7 marks)

III.

a) State a difference between perfect binary tree and full binary tree. (2 marks)
b) Draw a complete binary tree with at least 9 nodes (4 marks)

IV.

a) Draw a binary tree structure to insert the values given below.
12,23,15,8,7,11,14,18,16,24 (5 marks)

b) State two advantages of using tree data structure (2 marks)

c) Graphically represent the implementation of the binary tree (3 marks)

[Total 25 marks]

Question 5

I. State a difference between searching and sorting. (2 marks)

II.

a) Mention the steps to sort the numbers given below using selection sort algorithm.
11,12,25,7,20,22,13,24 (6 marks)

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b) What would be the output of the code segment given below. (4 marks)

```
#include <iostream>
using namespace std;
 void displayArray(int *a, int n)
{
 int i;
 for (i=0; i<n; i++)
 cout<<a[i]<<" ";
}
void swap(int *x,int *y)
{
 int t;
 t=(*x);
 *x=(*y);
 *y=t;
}
void function (int *a, int n)
{
 int i, j, m, t;
 for (i = 0; i < n-1; i++)
 {
 m = i;
 for (j = i+1; j < n; j++)
if (a[j] < a[m]) m = j;
 swap(&a[i],&a[m]);
 }
}
int main ()
{

 int a[] = {-10,-15,4,9,2,10,17,3,2,5};
 int n = 10;
 displayArray(a,n);cout<<endl;
 function(a, n);
 displayArray(a,n);
}
```

III. Write Pseudo code segment to represent the bubble sort algorithm (5 marks)

IV. Clearly mention the steps to retrieve the 56 from the list given below using binary search algorithm.

17,23,33,38,56,72

(8 marks)

[Total 25 marks]

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